

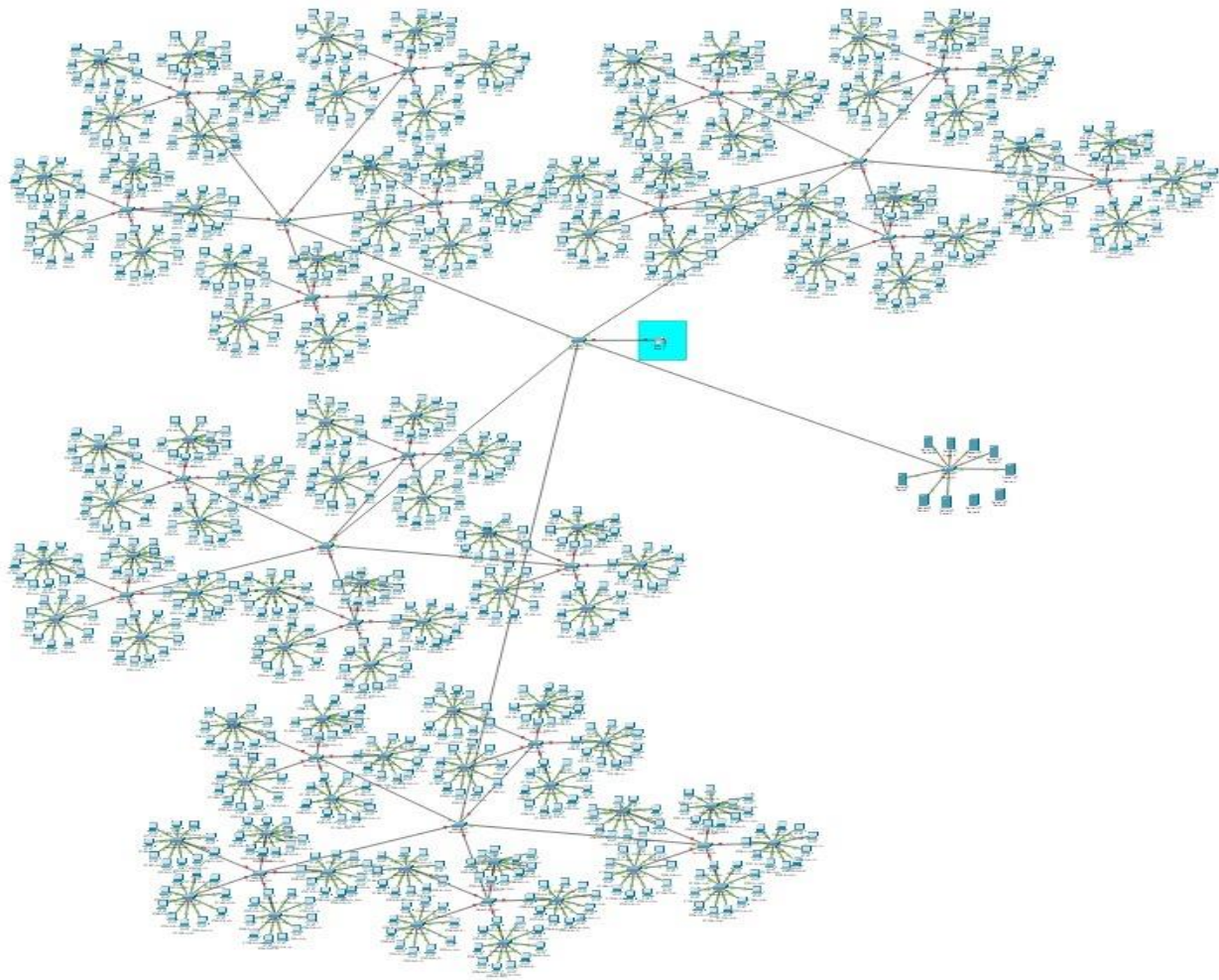
University Campus Network Design Project

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Introduction:

This project presents the design and implementation of a University Campus Network for a small university. The main objective is to build a scalable, secure, and organised network that connects multiple faculties, laboratories, and a Scientific Computation Center, while ensuring proper IP addressing, VLAN configuration, and network testing.

The network is designed using a hierarchical architecture that reflects campus networks. Private IP addressing, subnetting, VLAN configuration, and routing concepts are applied to achieve efficient communication and logical separation between different network segments where VLANs are extended across all floors to allow communication between devices that belong to the same VLAN.



Project Overview and Network Connection Scenario:

The university campus consists of the following main components:

- Four Faculties:

Each faculty is composed of:

4 floors where each floor contains 5 labs and each lab contains at least 10 PCs.

- PCs in each lab are connected to a Lab Switch
- Lab switches on the same floor are connected to a Floor Switch
- Floor switches are connected to a Faculty Switch

- One Scientific Computation Center (SCC):

Contains multiple servers where all servers are connected to a dedicated switch.

Faculty switches and the SCC switch are connected to the Main University Router. The main router provides connectivity between all faculties, the SCC, and the external network (Internet). This design follows a hierarchical architecture that improves scalability, performance, and fault isolation.

Network Topology and Connection Method:

The network uses a hierarchical star topology:

Access Layer: PCs connected to lab switches

Distribution Layer: Floor and faculty switches

Core Layer: Main university router

Trunk links are configured between lab switches, floor switches, and faculty switches to allow VLAN traffic to pass across different floors.

IP Addressing and Subnetting:

- Main Network Address:

The private IP address space used for the university campus is: 172.16.0.0 /16

NID: 172.16.0.0 /16

BID: 172.16.255.255 /16

This network is subnetted to serve each faculty and the Scientific Computation Center.

- Subnet Assignment: **Subnneting the mother network:**
 - **Bits Borrowed:** Determined the minimum number of bits required to create at least five subnets ($2^n \geq 5$), resulting in 3 bits being borrowed from the host portion.
 - **New Prefix Length:** The original /16 prefix was extended to /19 (16 + 3) to accommodate the new subnets.
 - **Subnet Increment:** The subnet IDs increment in the third octet by a block size of 32, derived from the value of the first borrowed bit (2^5).
 - **Total Subnets:** The borrowing of 3 bits creates a total of 8 available subnets, ensuring enough capacity for the 5 required faculties/labs.

After Subnetting calculations there would be total of eight subnets available, five of which are currently assigned, while the remaining subnets are reserved for future expansion.

	NID	Host IP range	BID
<i>Faculty 1</i>	172.16.0.0/19	172.16.0.1 – 172.16.31.254	172.16.31.255/19
<i>Faculty 2</i>	172.16.32.0/19	172.16.32.1 – 172.16.63.254	172.16.63.255/19
<i>Faculty 3</i>	172.16.64.0/19	172.16.64.1 – 172.16.95.254	172.16.95.255/19
<i>Faculty 4</i>	172.16.96.0/19	172.16.96.1 – 172.16.127.254	172.16.127.255/19
<i>SCC</i>	172.16.128.0/19	172.16.128.1 – 172.16.159.254	172.16.159.255/19
<i>Extra</i>	172.16.160.0/19	172.16.192.1 – 172.16.223.254	172.16.191.255/19
<i>Extra</i>	172.16.192.0/19	172.16.192.1 – 172.16.223.254	172.16.223.255/19
<i>Extra</i>	172.16.224.0/19	172.16.224.1 – 172.16.255.254	172.16.255.255/19

- PCs IP Assignment:

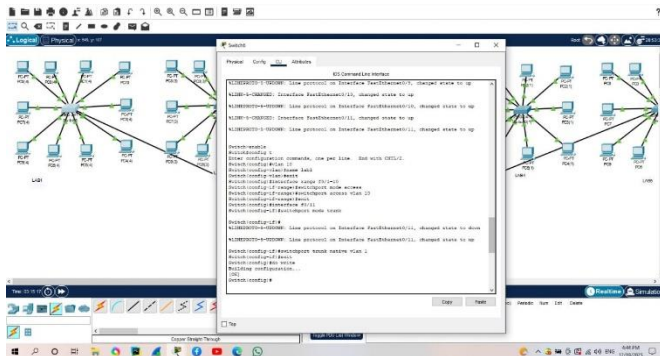
Each PC is assigned a static IP address within its lab subnet. The default gateway for each PC is configured as the faculty router interface.

VLAN Configuration:

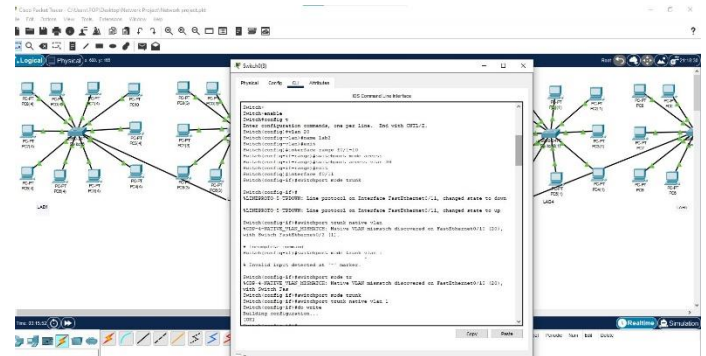
VLANs are configured based on lab numbers across all floors and faculties as required:

Lab number	VLAN ID
Lab 1	VLAN 10
Lab 2	VLAN 20
Lab 3	VLAN 30
Lab 4	VLAN 40
Lab 5	VLAN 50

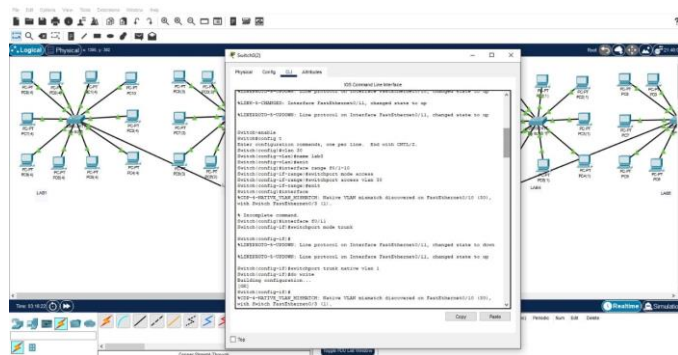
All labs with the same lab number belong to the same VLAN, even if they are located on different floors. VLANs are extended across the network using trunk links, which allows devices in the same VLAN to communicate successfully.



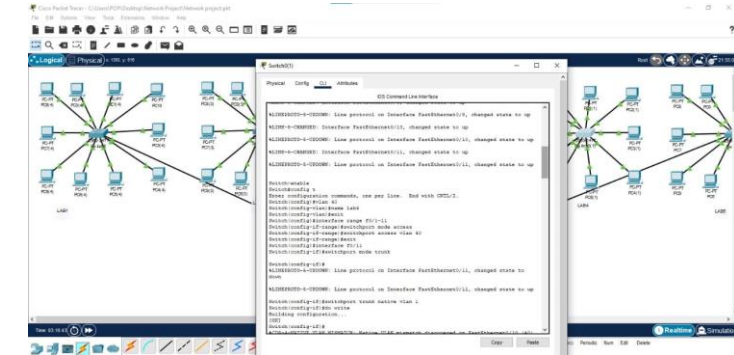
(Lab 1 switch configuration showing access ports assigned to VLAN 10)



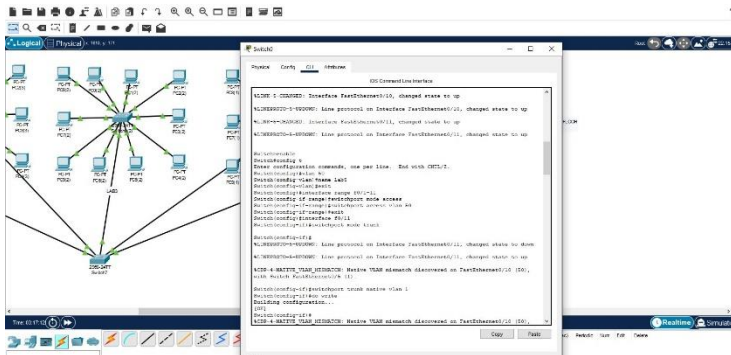
(Lab 2 switch configuration showing access ports assigned to VLAN 20)



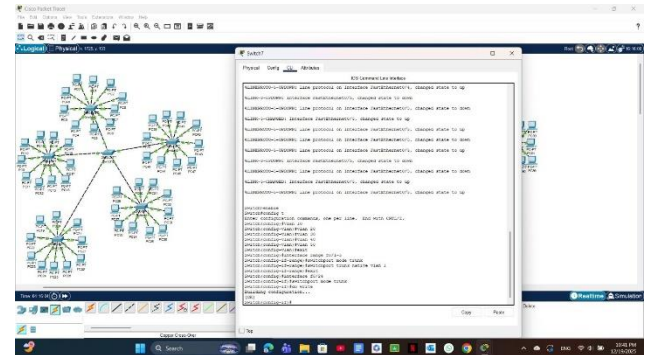
(Lab 3 switch configuration showing access ports assigned to VLAN 30)



(Lab 4 switch configuration showing access ports assigned to VLAN 40)



(Lab 5 switch configuration showing access ports assigned to VLAN 50)



(Floor Switch configuration showing VLAN creation and trunk port configuration)

Testing Scenarios:

Scenario 1:

Test Description: PC 1 on the first floor in the second lab in the Faculty of Computers and Artificial Intelligence (Faculty 1) accesses Server 1 in the Scientific Computation Center.

Source IP: 172.16.0.11

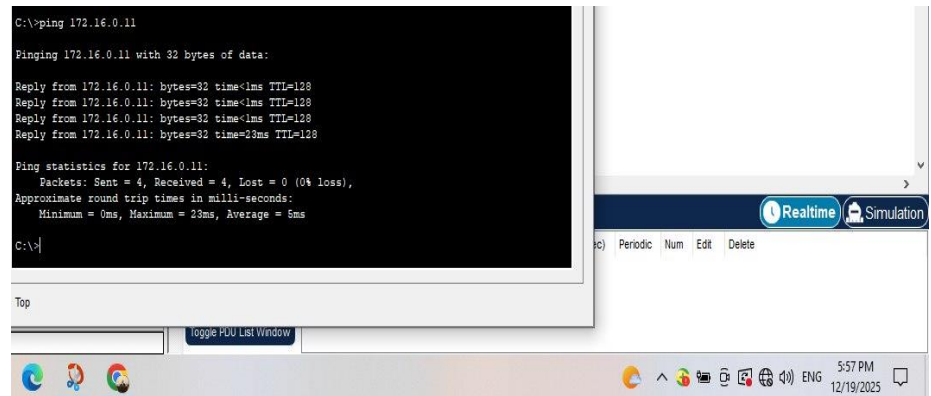
Destination IP: 172.16.128.1

Result: Successful communication due to correct routing and IP configuration.

Scenario 2:

Test Description: Test whether all labs that belong to the same VLAN can communicate with each other.

Expected Result: Successful communication because devices in the same VLAN are in the same broadcast domain.

A screenshot of a Windows command prompt window. The command entered is 'C:\>ping 172.16.0.11'. The output shows four successful replies from 172.16.0.11 with 32 bytes of data, each taking less than 1ms. The ping statistics show 4 packets sent, 4 received, 0 lost (0% loss), and approximate round trip times of 0ms minimum, 23ms maximum, and 5ms average. The window is titled 'C:\>' and has a 'loggie PDU List Window' tab open in the background.

(Network connectivity test results using the ping command to 172.16.0.11.)

Scenario 3:

Test Description: Test whether labs in different VLANs can communicate with each other.

Expected Result: Communication fails because inter-VLAN routing is not allowed without proper routing configuration.

Scenario 4:

Test Description: A PC in Faculty 1 attempts to communicate with a PC in Faculty 3.

Result: Successful communication through the main university router.

Scenario 5:

Test Description: A server in the Scientific Computation Center communicates with a PC in VLAN 10.

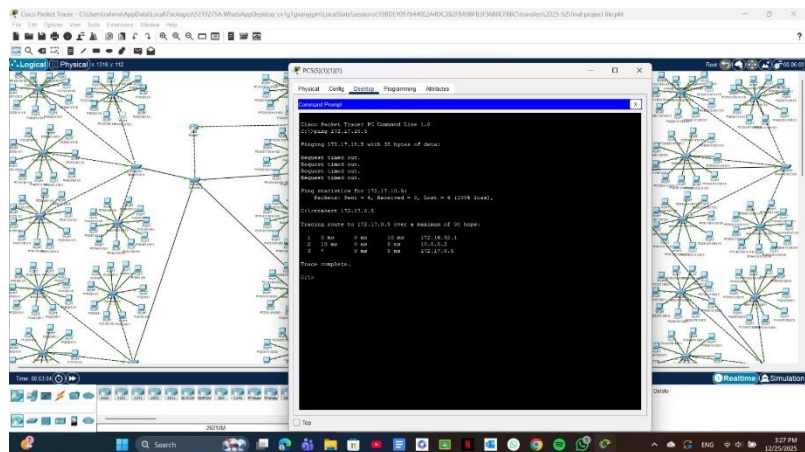
Result: Successful communication since routing is enabled between SCC and faculties.

Result: Communication fails, proving the importance of trunk links for VLAN traffic.

To enable efficient and scalable communication between Faculty 1 and Faculty 2, the OSPF (Open Shortest Path First) dynamic routing protocol was implemented on their respective faculty routers. This ensures that the routers can automatically learn and update paths to different subnets without manual static route entries.

Faculty 2: 2.2.2.2

Adjacency Establishment: Trunk links and router interfaces were monitored to ensure OSPF “Hello” packets established neighbor relationships.



(a traceroute command verifying that packets successfully travel from the source through the Faculty 2 gateway at 172.16.32.1 to reach the destination IP 172.17.0.5.)